

Venkat Dasari

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EDUCATION

University of Massachusetts Amherst

Master of Science in Data Analytics and Computational Social Science

Amherst, MA

September 2024 – May 2026

- **Thesis:** Analysis of Epistemic Evidentiality in American Political Writings

Gandhi Institute of Technology and Management

Bachelor of Technology in Computer Science and Engineering

Visakhapatnam, India

August 2019 – June 2023

EXPERIENCE

Research Consultant

Institute for Social Science Research

October 2025 – Present

Amherst, MA

- Deliver presentations and one-on-one consultations to researchers on quantitative text analysis, supporting social science projects from initial design through implementation in **R and Python**
- Develop tutorials and documentation for computational text analysis, covering **text preprocessing, EDA, topic models, and advanced AI/ML-based NLP techniques**

Team Lead

Gen AI Global

September 2025 – Present

Remote

- Define roles and lead a lean Architectural Team to advance the **AI 2027 initiative**, focusing on how AI and agentic architectures are designed and governed
- Assess risks across **AI/agentic architectures (model design, capability, interpretability)** and map technical controls and mitigations to the **NIST AI RMF**, producing traceable risk registers and implementation guidance for stakeholders
- Collaborate with legal leads to draft legislation and regulatory language that balance innovation with safety, aligning technical practices with emerging legal standards

Teaching Assistant

University of Massachusetts Amherst

September 2025 – Present

Amherst, MA

- Assisted in teaching **DACSS 603: Introduction to Quantitative Analysis**, covering statistics, probability, and regression analysis in R
- Held office hours and provided one-on-one tutoring sessions to **60+ students**, assisting with R (syntax, debugging, data wrangling, visualization, and regression) to support assignments and projects

Research Assistant

University of Massachusetts Amherst

November 2024 – Present

Amherst, MA

- Review literature and design computational frameworks to identify core values and citations in political books
- Implement **LLM-driven NLP pipelines (Python, Ollama, GPT, Llama)** to measure **500 books** for political values, citations, sentiments and ideological classifications
- Apply statistical methods including **latent class analysis (LCA)** and **Krippendorff's alpha** to validate constructs and measure consistency
- Execute Python scripts to collect and validate metadata for **500 books and 329 authors**, and develop an **interactive React dashboard** with faceted filters to visualize corpus statistics
- Build a public research portal for the **Ideological Books Corpus (IBC)** using **WordPress, HTML, CSS, and JavaScript**, publishing datasets, documentation, and updates to make the corpus accessible to scholars and public

Operational & Technical Assistant

The Bharat Scouts & Guides

January 2024 – June 2024

Rajahmundry, India

- Generated reports to help with campaign decision-making, resulting in an **increase of member count by 125% within 3 months**
- Created, managed and updated more than **800 member profiles** and members data
- Organized cultural events, social service and adventure programmes, in collaboration with Indian railways and other public organizations

Engineer Intern

Tata Elxsi

February 2023 – June 2023

Thiruvananthapuram, India

- Designed and developed a **React UI** for a resume parsing application, improving **user experience (UX)** and streamlining HR and project management workflows
- Identified parsing issues and implemented error and exception handling, increasing application reliability and processing efficacy by **20%**
- Researched embedded product engineering and communication protocols for automotive applications and delivered daily presentations to the team

TECHNICAL SKILLS

Languages: Python, R, C++, SQL, JavaScript, HTML, CSS

Tools & Technologies: vLLM, Ollama, Scikit-learn, Keras, TensorFlow, Flask, Node.js, React.js, Express.js, Apache Airflow, dbt (data build tool), Docker, Git

Databases: MySQL, MongoDB, Snowflake

Interests: Data Science, Machine Learning, Artificial Intelligence, Natural Language Processing, Data Engineering, Web Development, Research Computing

PROJECTS

Analysis of Epistemic Evidentiality in American Political Writings September 2025 – Present

- Deploy local Llama, Phi and Gemma models, and build parallelized inference pipelines on HPC cluster to conduct prompt engineering and fine-tuning (LoRA/PEFT) experiments
- Perform argument, topic, and evidentiality classification on 50 American political books using a human-in-the-loop (HITL) active learning workflow with LLMs; design evaluation metrics to quantify model uncertainty and false-confidence rates

Network Modeling of Post-Cold War Global Diplomacy[\(Link\)](#) February 2025 – May 2025

- Constructed a global diplomatic network for 140 countries (1995–2005) using Correlates of War project data and applied Temporal Exponential Random Graph Models (TERGM) to model diplomatic tie formation
- Identified strong reciprocity, transitivity, and decentralization in post-Cold War networks, signaling a shift toward interest-driven, multipolar international relations

Anti-Phish: Phishing Detection using Machine Learning[\(Link\)](#) August 2022 – May 2023

- Extracted 20+ features from 10,000 website URLs to build a comprehensive dataset and conducted a comparative study of supervised learning algorithms
- Implemented an XGBoost-based model with 95% classification accuracy and integrated the classifier into a Flask-based web application, enabling real-time phishing detection for end users

PUBLICATIONS

- S. S. P. Naresh, R. Talasu, **D. S. S. P. Venkat**, S. K. Korada and B. K. Mohanta, "Real Time Vehicle Tracking using YOLO Algorithm," 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-5, doi: 10.1109/ICCCNT56998.2023.10307265.

CERTIFICATIONS

- [Human Research — CITI Program \(Group 2: Social and Behavioral Research Investigators and Key Personnel\)](#)
- [Machine Learning with Python — IBM](#)
- [Introduction to Artificial Intelligence — IBM](#)
- [Software Development Processes and Methodologies — University of Minnesota](#)
- [Technical Support Fundamentals — Google](#)

ACHIEVEMENTS & EXTRACURRICULARS

- Volunteered as a **Data Consultant at Five College DataFest 2025 (American Statistical Association)**, assisting student teams with statistical guidance, R/Python troubleshooting, and presentation feedback
- Secured an **All India Rank (AIR) of 20** in the **National Engineering Olympiad 6.0**
- Represented Hex Esports, Minus 40 Club, and other organizations in PUBG Mobile tournaments as an academic esports athlete and finalist in four national-level competitions
- Actively participated in The Bharat Scouts and Guides, developing teamwork, leadership, and life skills through service and adventure activities